Loan Management System (LMS) – End-to-End Business Analysis Case Study

Prepared by: Business Analyst | **Client (fictional):** Meridian Finance Ltd. (retail NBFC) | **Version:** 1.0 | **Date:** November 2026

> *Note: Meridian Finance is a fictional organisation. All figures are illustrative assumptions made for portfolio demonstration.*

1. Executive Summary

Meridian Finance disburses personal, two-wheeler and small-business loans through branches and a DSA (agent) network. Applications, KYC, credit checks, approvals and collections run across spreadsheets, email and a legacy desktop tool. This causes slow turnaround, errors, weak collections visibility and audit risk.

This case study defines a **Loan Management System (LMS)** that digitises the loan lifecycle: *Apply → Verify → Underwrite → Approve → Disburse → Repay → Collect → Close*. It covers requirements (BRD/FRD), user stories, process redesign, wireframes, data/analytics (SQL + Power BI) and UAT.

2. Business Problem and Objectives

2.1 Problem Statement

Loan processing at Meridian is manual and fragmented, resulting in long approval times, high drop-off, inconsistent credit decisions, delayed collections and limited portfolio visibility for management.

2.2 Current Pain Points (Baseline)

#	Pain Point	Baseline (illustrative)	Impact
1	Application-to-disbursal turnaround (TAT)	9 working days	Customers drop to competitors
2	Application drop-off before decision	38%	Lost revenue

#	Pain Point	Baseline (illustrative)	Impact
3	Data-entry/rework errors in applications	14% of files	Rework cost, delays
4	Manual EMI tracking & reminders	Spreadsheet based	Missed follow-ups
5	Portfolio at Risk (PAR 30)	11.5%	Credit loss, provisioning
6	Collection efficiency	88%	Cash-flow pressure
7	MIS report preparation	3 days/month, manual	Late decisions
8	Audit findings on KYC/document gaps	22 per year	Regulatory risk

2.3 Business Objectives (SMART)

ID	Objective	Target	Timeline
BO-1	Reduce application-to-disbursal TAT	9 days → 3 days	6 months post go-live
BO-2	Reduce application drop-off	38% → 20%	6 months
BO-3	Reduce data-entry errors	14% → <3%	3 months
BO-4	Improve collection efficiency	88% → 95%	9 months
BO-5	Reduce PAR 30	11.5% → 8%	12 months
BO-6	Real-time MIS (replace manual reports)	3 days → real time	Go-live
BO-7	Zero critical KYC/audit gaps	22 → 0	12 months

2.4 Scope

In scope: Customer & application management, KYC/document management, credit scoring & underwriting workflow, approval matrix, disbursement, EMI schedule & repayment, collections & reminders, closure/foreclosure, reporting dashboards, role-based access and audit trail.

Out of scope (Phase 1): Co-lending, secondary-market securitisation, mobile app for customers (web portal only), general-ledger replacement (integration only).

2.4 Assumptions and Constraints

- Credit bureau, e-KYC and payment-gateway APIs are available via vendors.
- Compliance with RBI digital lending guidelines and data-privacy norms is mandatory.
- Go-live budget and timeline: 9 months, phased release.

3. Stakeholder Analysis

3.1 Stakeholder Register

Stakeholder	Role	Interest / Need	Influence	Interest	Engagement
COO (Sponsor)	Executive sponsor	TAT, cost, growth	High	High	Manage closely
Chief Risk Officer	Credit policy owner	Risk controls, PAR	High	High	Manage closely
Head of Lending Ops	Process owner	Efficiency, SLAs	High	High	Manage closely
Credit Underwriters	Primary user	Clear data, scorecards	Medium	High	Involve (workshops)
Relationship Managers / DSAs	Primary user	Fast onboarding, status visibility	Medium	High	Involve
Collections Team	Primary user	Prioritised worklists	Medium	High	Involve
Compliance & Audit	Control owner	KYC, audit trail, RBI rules	High	Medium	Keep satisfied
IT / Enterprise Architecture	Delivery & security	Integration, security, support	Medium	Medium	Keep informed/consult
Finance & Accounts	Downstream user	Accurate disbursement/receipts	Medium	Medium	Keep informed
Customers (borrowers)	End user	Quick, transparent loans	Low	High	Keep informed (surveys, UAT sample)
Credit Bureau / Payment Gateway	External vendors	Stable API usage	Low	Low	Monitor

3.2 Power–Interest Grid

- **Manage closely:** COO, CRO, Head of Lending Ops
- **Keep satisfied:** Compliance & Audit
- **Keep informed:** IT, Finance, Customers
- **Involve as SMEs:** Underwriters, RMs/DSAs, Collections

3.3 RACI (Key Deliverables)

Deliverable	BA	Sponsor	Ops Head	CRO	IT	Compliance
BRD	R	A	C	C	C	C
FRD	R	I	C	C	C	C
Credit rules/approval matrix	C	I	C	A/R	I	C
Solution build	C	I	I	I	A/R	I
UAT sign-off	R	A	R	R	C	R

R = Responsible, A = Accountable, C = Consulted, I = Informed

4. Business Requirements Document (BRD)

4.1 Business Requirements (MoSCoW)

ID	Business Requirement	Objective	Priority
BR-01	Capture loan applications digitally via branch, DSA and web portal	BO-1, BO-2	Must
BR-02	Automate KYC (e-KYC/PAN/Aadhaar-consent) and document collection	BO-1, BO-7	Must
BR-03	Integrate credit bureau and generate an automated credit score	BO-1, BO-5	Must
BR-04	Configurable underwriting rules and tiered approval matrix	BO-5, BO-7	Must
BR-05	Auto-generate sanction letter, agreement and EMI schedule	BO-1	Must
BR-06	Disbursement with maker-checker control	BO-7	Must
BR-07	Repayment tracking, auto-allocation and receipt generation	BO-4	Must

ID	Business Requirement	Objective	Priority
BR-08	Automated reminders and prioritised collections worklist	BO-4, BO-5	Must
BR-09	Foreclosure, part-payment and loan closure with NOC	BO-7	Should
BR-10	Real-time dashboards for management and operations	BO-6	Must
BR-11	Role-based access, full audit trail, data encryption	BO-7	Must
BR-12	Customer self-service portal (status, statement, pay EMI)	BO-2	Should
BR-13	Early-warning alerts for at-risk accounts	BO-5	Could
BR-14	WhatsApp notification channel	BO-2	Could
BR-15	Co-lending and securitisation	–	Won't (Phase 1)

4.2 Success Metrics (KPIs)

TAT, approval rate, drop-off rate, disbursement volume, collection efficiency, PAR 30/60/90, NPA %, cost per loan, UAT defect leakage, user adoption (>90%).

4.3 Risks and Mitigations

Risk	Likelihood	Impact	Mitigation
Vendor API downtime (bureau/e-KYC)	Medium	High	Fallback queue, retry logic, SLA with vendors
User resistance to change	High	Medium	Early SME involvement, training, champions
Regulatory change mid-project	Medium	High	Configurable rules, compliance review gates
Poor legacy data quality	High	Medium	Data cleansing and migration validation
Scope creep	Medium	Medium	Change control board, MoSCoW discipline

5. Functional Requirements Document (FRD)

5.1 Functional Requirements

ID	Module	Functional Requirement	Traces to
FR-01	Application	System shall capture applicant, employment, income, loan amount, tenure and product; mandatory fields validated on submit	BR-01
FR-02	Application	System shall auto-generate unique Application ID (format LN-YYYY-NNNNNN)	BR-01
FR-03	Application	System shall detect duplicate applicants using PAN + mobile	BR-01
FR-04	KYC	System shall verify PAN and Aadhaar via e-KYC API and store the consent record	BR-02
FR-05	KYC	System shall allow upload of PDF/JPG/PNG up to 5 MB per document and flag missing documents	BR-02
FR-06	Credit	System shall fetch bureau score and compute internal score using configured scorecard	BR-03
FR-07	Credit	System shall calculate FOIR and flag cases exceeding 50%	BR-04
FR-08	Underwriting	System shall route applications by approval matrix (see 5.2)	BR-04
FR-09	Underwriting	Underwriter shall approve, reject or refer back with mandatory reason code	BR-04
FR-10	Sanction	System shall generate sanction letter and loan agreement from templates	BR-05
FR-11	Sanction	System shall generate EMI schedule (reducing balance)	BR-05
FR-12	Disbursement	Disbursement shall require maker-checker approval and bank-account validation (penny drop)	BR-06
FR-13	Repayment	System shall record payments (UPI, NACH, cheque, cash), allocate in order: penalty → interest → principal	BR-07
FR-14	Repayment	System shall generate receipt and update outstanding balance in real time	BR-07

ID	Module	Functional Requirement	Traces to
FR-15	Collections	System shall send SMS/email reminders at T-3, T-0 and T+3 days	BR-08
FR-16	Collections	System shall create a prioritised worklist by DPD bucket and outstanding amount	BR-08
FR-17	Closure	System shall compute foreclosure amount, accept part-payment and issue NOC on closure	BR-09
FR-18	Reporting	System shall provide dashboards and scheduled exports (Excel/PDF)	BR-10
FR-19	Security	System shall enforce role-based access, log every create/update/approve action with user and timestamp	BR-11

5.2 Business Rules

Rule	Description
BRL-01	Minimum age 21, maximum age at maturity 60; minimum bureau score 650 for auto-processing
BRL-02	FOIR (existing EMIs + proposed EMI ÷ net monthly income) must not exceed 50%
BRL-03	Approval matrix: up to ₹5L – Underwriter; ₹5L–₹15L – Senior Underwriter; above ₹15L – Credit Committee
BRL-04	Score ≥ 750 and FOIR ≤ 40% – eligible for straight-through approval (up to ₹3L)
BRL-05	Penal interest 2% per month on overdue EMI; configurable by product
BRL-06	DPD buckets: 0, 1–30, 31–60, 61–90, 90+ (NPA)
BRL-07	Sanction validity: 30 days from issue

5.3 EMI Formula

`EMI = P × r × (1+r)^n / ((1+r)^n – 1)` where P = principal, r = monthly rate, n = tenure in months.

5.4 Application Status Lifecycle

`Draft → Submitted → KYC Pending → Under Review → Referred Back → Approved / Rejected → Sanctioned → Disbursed → Active → Closed / NPA`

5.5 Non-Functional Requirements

ID	Category	Requirement
NFR-01	Performance	Page response < 3 sec for 95% of requests; support 500 concurrent users
NFR-02	Availability	99.5% uptime; RPO ⩽ 15 min, RTO ⩽ 4 hrs
NFR-03	Security	AES-256 encryption at rest, TLS 1.2+ in transit, MFA for admin roles
NFR-04	Compliance	Audit logs retained 8 years; RBI digital lending and data-privacy norms
NFR-05	Usability	Responsive web, WCAG 2.1 AA, English plus one regional language
NFR-06	Scalability	Handle 3× current volume without redesign

5.6 Core Data Entities

Customer, Loan Product, Loan Application, Document, Credit Assessment, Loan Account, Repayment Schedule, Payment, Collection Activity, User/Role, Audit Log. *(Schema in Section 9.)*

6. User Stories and Acceptance Criteria

6.1 Epics

Epic	Name	Stories
E1	Loan Origination	US-01, US-02
E2	KYC & Credit Assessment	US-03, US-04
E3	Approval & Disbursement	US-05, US-06
E4	Repayment & Collections	US-07, US-08
E5	Reporting & Governance	US-09, US-10

6.2 Stories

US-01 – Submit loan application *(FR-01, FR-02, FR-03 | Must | 5 pts)* As a **Relationship Manager**, I want to capture an application digitally so that the customer is onboarded in minutes without paperwork.

- **AC1:** *Given* all mandatory fields are filled, *when* I click Submit, *then* the system creates an Application ID and sets status to "Submitted".
- **AC2:** *Given* a mandatory field is blank, *when* I click Submit, *then* the field is highlighted with an error and submission is blocked.
- **AC3:** *Given* the PAN and mobile already exist on an active application, *when* I submit, *then* a duplicate warning is shown with a link to the existing record.

US-02 – Save draft and resume *(FR-01 | Should | 3 pts)* As an **RM**, I want to save an incomplete application so that I can resume later.

- **AC1:** *Given* partial data, *when* I click Save Draft, *then* the application is saved with status "Draft" and no validation errors for optional fields.
- **AC2:** *Given* a draft older than 15 days, *then* the system auto-archives it and notifies the RM.

US-03 – Verify KYC electronically *(FR-04, FR-05 | Must | 8 pts)* As a **Loan Processor**, I want KYC verified automatically so that I avoid manual document checks.

- **AC1:** *Given* a valid PAN and Aadhaar consent, *when* verification runs, *then* status shows "KYC Verified" and the response is stored.
- **AC2:** *Given* the API fails or times out, *then* the case moves to "KYC Pending", is retried 3 times, and an alert is raised to the processor.

- **AC3:** *Given* a document exceeds 5 MB or has an unsupported format, *then* upload is rejected with a clear message.

US-04 – View credit score and FOIR *(FR-06, FR-07 | Must | 8 pts)* As an **Underwriter**, I want bureau score, internal score and FOIR in one view so that I can decide quickly.

- **AC1:** *Given* a KYC-verified application, *when* I open it, *then* bureau score, internal score, FOIR and existing obligations are displayed.
- **AC2:** *Given* FOIR > 50%, *then* a red "Policy Breach" flag is displayed and Approve is disabled unless an exception is approved.

US-05 – Approve, reject or refer back *(FR-08, FR-09 | Must | 8 pts)* As an **Underwriter**, I want to take a decision with a reason code so that decisions are consistent and auditable.

- **AC1:** *Given* a loan within my approval limit, *when* I approve, *then* status becomes "Approved" and the sanction letter is generated.
- **AC2:** *Given* a loan above my limit, *then* the system routes it automatically to the next approver level.
- **AC3:** *Given* I reject or refer back, *when* no reason code is selected, *then* the action is blocked.

US-06 – Disburse loan with maker-checker *(FR-12 | Must | 5 pts)* As a **Disbursement Officer**, I want a second-level check before funds are released so that fraud risk is reduced.

- **AC1:** *Given* a maker has initiated disbursement, *then* the same user cannot approve it as checker.
- **AC2:** *Given* penny-drop validation fails, *then* disbursement is blocked and the reason is displayed.
- **AC3:** *Given* successful disbursement, *then* status becomes "Active" and the EMI schedule is locked.

US-07 – Record repayment *(FR-13, FR-14 | Must | 5 pts)* As a **Collections Executive**, I want payments posted and allocated automatically so that balances are always accurate.

- **AC1:** *Given* a payment is received, *then* it is allocated to penalty, then interest, then principal, and a receipt is generated.
- **AC2:** *Given* a payment exceeds the due amount, *then* the excess reduces principal and the schedule is recalculated.

US-08 – Prioritised collections worklist *(FR-15, FR-16 | Must | 5 pts)* As a **Collections Manager**, I want a daily worklist sorted by DPD and amount so that my team focuses on high-risk accounts.

- **AC1:** *Given* it is 8:00 AM, *then* the worklist is generated with accounts grouped by DPD bucket.
- **AC2:** *Given* an agent logs a call outcome or promise-to-pay date, *then* it is saved and a follow-up task is created.

US-09 – Management dashboard *(FR-18 | Must | 8 pts)* As a **COO**, I want a real-time dashboard so that I can monitor disbursement, TAT and portfolio quality.

- **AC1:** *Given* I open the dashboard, *then* KPIs refresh from data no older than 15 minutes.
- **AC2:** *Given* I filter by branch, product or month, *then* all visuals update consistently.

US-10 – Audit trail *(FR-19 | Must | 3 pts)* As a **Compliance Officer**, I want an immutable log of all actions so that I can respond to audits.

- **AC1:** *Given* any create/update/approve action, *then* user, timestamp, old value and new value are logged.
- **AC2:** *Given* a user tries to edit an audit record, *then* the action is denied.

7. AS-IS and TO-BE Process Flows

7.1 AS-IS Process (Manual)

Step	Actor	Activity	Problem
1	Customer / DSA	Fills paper form, submits photocopies	Incomplete forms, loss of documents
2	Branch staff	Re-types data in Excel/legacy tool	Duplicate entry, typos (14% error)
3	Back-office	Manually verifies KYC, emails scans	1–2 days delay
4	Credit analyst	Pulls bureau report manually, calculates FOIR in Excel	Inconsistent decisions
5	Approver	Reviews physical file / email chain	Approval delay of 2–3 days
6	Operations	Prepares sanction letter in Word, sends for signature	Manual, error-prone
7	Finance	Disburses after emailed approval	Weak maker-checker control
8	Collections	Tracks EMIs in spreadsheets, calls customers	Missed follow-ups, 88% efficiency

Step	Actor	Activity	Problem
9	Management	Requests monthly MIS from ops	3 days of manual effort

Pain summary: ~9-day TAT, no real-time status, audit gaps, reactive collections.

7.2 TO-BE Process (LMS-Enabled)

Step	Actor	Activity	Improvement
1	Customer / RM / DSA	Submits digital application with e-documents	Single entry, validated at source
2	System	Duplicate check, e-KYC, bureau pull, scorecard and FOIR auto-calculated	Minutes instead of days
3	System	Rules engine: auto-approve / auto-reject / route for manual review	Straight-through processing for eligible cases
4	Underwriter	Reviews exceptions on one consolidated screen; approves, rejects or refers back	Consistent, policy-driven decisions
5	System	Generates sanction letter and agreement; customer e-signs	No paper
6	Disbursement maker/checker	Penny-drop validation and dual authorisation	Fraud control
7	System	Creates loan account, EMI schedule and e-NACH mandate	Automated
8	System	Sends reminders and creates prioritised collections worklists	Proactive collections
9	Management	Views real-time Power BI dashboard	Instant visibility

7.3 Process Flow (Text Diagram)

```
[Start] → Digital Application → Duplicate Check → e-KYC
   → Bureau Score + FOIR → Rules Engine
        ├─ Score ≥ 750 & FOIR ≤ 40% & ≤ ₹3L → Auto-Approve ─┐
        ├─ Policy fail → Auto-Reject → Notify customer → [End]|
        └─ Else → Underwriter Review (per approval matrix)    |
              ├─ Approve ──────────────────────────────────────┤
              ├─ Refer Back → RM updates → re-enter review    |
              └─ Reject → Notify customer → [End]             |
                                                              ↓
   Sanction Letter + E-Sign → Maker-Checker Disbursement → Active Loan
   → EMI Collection / Reminders → [Closed] or [Collections → Recovery]
```

7.4 Gap Analysis

Area	AS-IS	TO-BE	Benefit
TAT	9 days	3 days	−67%
Data entry	Multiple, manual	Single, validated	Errors < 3%
Credit decision	Analyst judgement	Rule + scorecard driven	Consistency
Disbursal control	Email approval	Maker-checker	Fraud reduction
Collections	Reactive	Predictive, prioritised	+7 pts efficiency
MIS	Monthly, manual	Real-time	Faster decisions

8. Wireframes

Low-fidelity wireframes for the three highest-value screens. Annotations in [brackets] map to functional requirements.

8.1 WF-01: New Loan Application (RM / DSA)

```
+-------------------------------------------------------------------+
| MERIDIAN LMS   > Applications > New Application     [User: RM_042] |
+-------------------------------------------------------------------+
| Step: (1) Applicant  (2) Employment  (3) Loan  (4) Documents  (5) Review |
+-------------------------------------------------------------------+
| APPLICANT DETAILS                                                  |
| Full Name*   [________________________]  DOB*  [DD/MM/YYYY]        |
| PAN*         [__________]  [Verify]    Mobile* [__________] [OTP]  |
```

```
| Email        [_____________________]                          |
| Address*     [_____________________________________________]   |
|                                                                |
| (!) Duplicate found for PAN+Mobile: LN-2026-004512 [View]  [FR-03] |
|                                                                |
| LOAN REQUEST                                                   |
| Product*  [Personal Loan v]  Amount* [₹ ________]  Tenure* [__] mo |
| Indicative EMI: ₹ ______  (auto-calculated)            [FR-11]   |
|                                                                |
|            [Save Draft]            [Back]   [Next >]           |
+----------------------------------------------------------------+
  * mandatory field   [FR-01, FR-02]
```

8.2 WF-02: Underwriter Workbench

```
+----------------------------------------------------------------+
| Application LN-2026-004871 | Status: UNDER REVIEW | SLA: 5h 20m left |
+----------------------------------------------------------------+
| APPLICANT SNAPSHOT        | CREDIT ASSESSMENT           [FR-06/07]|
| Name: A. Kumar            | Bureau Score:   712                  |
| Age: 34  Income: ₹85,000  | Internal Score: 68 / 100             |
| Employer: ABC Pvt Ltd     | Existing EMIs:  ₹22,000               |
| KYC: [Verified]           | Proposed EMI:   ₹14,500               |
| Loan: ₹4,00,000 / 36 mo   | FOIR:  43%   [  Within limit (<=50%) ]|
+----------------------------------------------------------------+
| DOCUMENTS: [PAN OK] [Aadhaar OK] [Salary slips OK] [Bank stmt (!)] |
+----------------------------------------------------------------+
| RULE CHECKS:  Age OK | Bureau >= 650 OK | FOIR OK | Policy flags: 0 |
+----------------------------------------------------------------+
| DECISION  ( ) Approve  ( ) Reject  ( ) Refer Back    [FR-09]       |
| Reason code* [Select v]      Remarks [_____________________]       |
| Approval level required: Senior Underwriter (₹5L-₹15L band) [FR-08]  |
|                          [Submit Decision]                      |
+----------------------------------------------------------------+
```

8.3 WF-03: Management Dashboard (Summary View)

```
+--------------------------------------------------------------------+
| LMS PORTFOLIO DASHBOARD    Filters: [Month v] [Branch v] [Product v]|
+--------------------------------------------------------------------+
| [Disbursed MTD]  [Avg TAT]   [Approval %]  [Coll. Eff.]  [PAR 30]  |
|   ₹ 48.2 Cr       3.4 days     71%           93.6%        9.1%      |
+--------------------------------------------------------------------+
| Monthly Disbursement (Column)  |  Application Funnel (Funnel)      |
|  |  ||  ||| ||||               |  Submitted 1000 -> Approved 710    |
|  +------------------------     |  -> Disbursed 640                  |
+---------------------------------+-----------------------------------+
| DPD Bucket Mix (Donut)         |  Branch Performance (Table/Bar)    |
|  0 | 1-30 | 31-60 | 61-90 | 90+|  Branch | Disb | PAR30 | TAT      |
+--------------------------------------------------------------------+
```

9. SQL Analysis

Dialect: SQLite (runs in DB Browser for SQLite). Minor changes needed for SQL Server/PostgreSQL (e.g., date functions).

9.1 Data Model

```
CREATE TABLE customers (
  customer_id INTEGER PRIMARY KEY, full_name TEXT, pan TEXT UNIQUE,
  city TEXT, monthly_income REAL, bureau_score INTEGER, created_date DATE);

CREATE TABLE loan_products (
  product_id INTEGER PRIMARY KEY, product_name TEXT,
  interest_rate REAL, max_tenure_months INTEGER);

CREATE TABLE loan_applications (
  application_id INTEGER PRIMARY KEY, customer_id INTEGER, product_id
INTEGER,
  branch TEXT, channel TEXT, requested_amount REAL, tenure_months INTEGER,
  status TEXT,            -- Submitted/Approved/Rejected/Disbursed
  application_date DATE, decision_date DATE, disbursal_date DATE,
  FOREIGN KEY (customer_id) REFERENCES customers(customer_id),
  FOREIGN KEY (product_id) REFERENCES loan_products(product_id));

CREATE TABLE loans (
  loan_id INTEGER PRIMARY KEY, application_id INTEGER,
  disbursed_amount REAL, outstanding_principal REAL,
  disbursal_date DATE, status TEXT);   -- Active/Closed/NPA
```

```
CREATE TABLE repayment_schedule (
  schedule_id INTEGER PRIMARY KEY, loan_id INTEGER, emi_no INTEGER,
  due_date DATE, emi_amount REAL, paid_amount REAL DEFAULT 0, paid_date
DATE);
```

9.2 Queries

Q1. Application funnel by status

```
SELECT status, COUNT(*) AS applications,
       ROUND(100.0 * COUNT(*) / SUM(COUNT(*)) OVER (), 1) AS pct_of_total
FROM loan_applications
GROUP BY status
ORDER BY applications DESC;
```

Q2. Approval rate and average TAT by product

```
SELECT p.product_name,
       COUNT(*) AS applications,
       ROUND(100.0 * SUM(CASE WHEN a.status IN ('Approved','Disbursed') THEN
1 ELSE 0 END)
             / COUNT(*), 1) AS approval_rate_pct,
       ROUND(AVG(julianday(a.disbursal_date) -
julianday(a.application_date)), 1) AS avg_tat_days
FROM loan_applications a
JOIN loan_products p ON p.product_id = a.product_id
GROUP BY p.product_name;
```

Q3. Monthly disbursement with month-over-month growth (LAG)

```
WITH monthly AS (
  SELECT strftime('%Y-%m', disbursal_date) AS month,
         SUM(disbursed_amount) AS disbursed
  FROM loans GROUP BY 1)
SELECT month, disbursed,
       LAG(disbursed) OVER (ORDER BY month) AS prev_month,
       ROUND(100.0 * (disbursed - LAG(disbursed) OVER (ORDER BY month))
             / LAG(disbursed) OVER (ORDER BY month), 1) AS mom_growth_pct
FROM monthly;
```

Q4. Overdue EMIs with Days Past Due and bucket

```
SELECT l.loan_id, c.full_name, r.emi_no, r.due_date,
       (r.emi_amount - r.paid_amount) AS overdue_amount,
       CAST(julianday('2026-10-31') - julianday(r.due_date) AS INTEGER) AS
dpd,
       CASE
         WHEN julianday('2026-10-31') - julianday(r.due_date) <= 30 THEN '1-
30'
         WHEN julianday('2026-10-31') - julianday(r.due_date) <= 60 THEN '31-
60'
         WHEN julianday('2026-10-31') - julianday(r.due_date) <= 90 THEN '61-
90'
         ELSE '90+' END AS dpd_bucket
FROM repayment_schedule r
JOIN loans l ON l.loan_id = r.loan_id
JOIN loan_applications a ON a.application_id = l.application_id
JOIN customers c ON c.customer_id = a.customer_id
WHERE r.due_date < '2026-10-31' AND r.paid_amount < r.emi_amount;
```

Q5. Portfolio at Risk (PAR 30)

```
WITH overdue_loans AS (
  SELECT DISTINCT loan_id
  FROM repayment_schedule
  WHERE paid_amount < emi_amount
    AND julianday('2026-10-31') - julianday(due_date) > 30)
SELECT ROUND(100.0 * SUM(CASE WHEN o.loan_id IS NOT NULL THEN
l.outstanding_principal ELSE 0 END)
             / SUM(l.outstanding_principal), 2) AS par_30_pct
FROM loans l
LEFT JOIN overdue_loans o ON o.loan_id = l.loan_id
WHERE l.status = 'Active';
```

Q6. Collection efficiency by month

```
SELECT strftime('%Y-%m', due_date) AS month,
       SUM(emi_amount) AS demand,
       SUM(paid_amount) AS collected,
       ROUND(100.0 * SUM(paid_amount) / SUM(emi_amount), 1) AS
collection_eff_pct
FROM repayment_schedule
WHERE due_date <= '2026-10-31'
GROUP BY 1 ORDER BY 1;
```

Q7. Branch ranking by disbursement (DENSE_RANK)

```
SELECT branch, SUM(requested_amount) AS total_disbursed,
       DENSE_RANK() OVER (ORDER BY SUM(requested_amount) DESC) AS branch_rank
FROM loan_applications
WHERE status = 'Disbursed'
GROUP BY branch;
```

Q8. Customer risk quartiles by bureau score (NTILE)

```
SELECT customer_id, full_name, bureau_score,
       NTILE(4) OVER (ORDER BY bureau_score DESC) AS risk_quartile
FROM customers;
```

Q9. Channel conversion (DSA vs Branch vs Web)

```
SELECT channel, COUNT(*) AS applications,
       SUM(CASE WHEN status = 'Disbursed' THEN 1 ELSE 0 END) AS disbursed,
       ROUND(100.0 * SUM(CASE WHEN status = 'Disbursed' THEN 1 ELSE 0 END) /
COUNT(*), 1) AS conversion_pct
FROM loan_applications
GROUP BY channel ORDER BY conversion_pct DESC;
```

10. Power BI Dashboard

10.1 Data Model (Star Schema)

- **Fact tables:** FactApplications, FactLoans, FactRepayments
- **Dimensions:** DimCustomer, DimProduct, DimBranch, DimChannel, DimDate (marked as date table)
- **Relationships:** Dim → Fact, one-to-many, single direction; DimDate related on application, disbursal and due dates (one active, others via USERELATIONSHIP)
- **Refresh:** Scheduled every 15 minutes via gateway (DirectQuery for FactRepayments if volume requires)

10.2 Key DAX Measures

```
Total Disbursed = SUM(FactLoans[DisbursedAmount])

Applications = COUNTROWS(FactApplications)

Approval Rate % =
DIVIDE(
    CALCULATE(COUNTROWS(FactApplications),
              FactApplications[Status] IN { "Approved", "Disbursed" }),
```

```
    [Applications])

Avg TAT (Days) =
AVERAGEX(
    FILTER(FactApplications, NOT ISBLANK(FactApplications[DisbursalDate])),
    DATEDIFF(FactApplications[ApplicationDate],
FactApplications[DisbursalDate], DAY))

Collection Efficiency % =
DIVIDE(SUM(FactRepayments[PaidAmount]), SUM(FactRepayments[EmiAmount]))

PAR 30 % =
DIVIDE(
    CALCULATE(SUM(FactLoans[OutstandingPrincipal]), FactLoans[MaxDPD] > 30),
    SUM(FactLoans[OutstandingPrincipal]))

Disbursed MoM % =
VAR Curr = [Total Disbursed]
VAR Prev = CALCULATE([Total Disbursed], DATEADD(DimDate[Date], -1, MONTH))
RETURN DIVIDE(Curr - Prev, Prev)
```

10.3 Report Pages

Page	Audience	Visuals	Insight Delivered
1. Executive Summary	COO, CRO	KPI cards (Disbursed, TAT, Approval %, Coll. Eff., PAR 30), monthly disbursement column + MoM line, funnel	Overall health and trend
2. Origination & TAT	Ops Head	Funnel, TAT by stage, channel conversion bar, rejection reasons	Where applications stall or drop off
3. Portfolio Quality	CRO	DPD bucket donut, PAR trend line, product-wise PAR, bureau-score band matrix	Credit risk concentration
4. Collections	Collections Mgr	Collection efficiency by branch, overdue table with drill-through, promise-to-pay tracker	Where to focus recovery effort

Design standards: consistent colour palette (green = on target, amber = watch, red = breach), slicers for Month/Branch/Product/Channel synced across pages, tooltips with definitions, row-level security by branch for branch managers.

11. UAT Test Scenarios

11.1 UAT Approach

- **Environment:** Production-like UAT with masked data and sandbox vendor APIs
- **Testers:** Business users (RMs, underwriters, processors, collections, finance, compliance)
- **Entry criteria:** SIT complete with no open critical/high defects; test data loaded; UAT scripts approved
- **Exit criteria:** 100% critical scenarios executed and passed; no open Severity 1/2 defects; ≥95% overall pass rate; business sign-off obtained

Severity	Definition
S1 – Critical	System unusable or financial/compliance impact; no workaround
S2 – High	Major function fails; workaround exists
S3 – Medium	Minor function issue
S4 – Low	Cosmetic

11.2 Test Scenarios

TC ID	Story / FR	Scenario	Test Steps (summary)	Expected Result	Type
UAT-01	US-01 / FR-01,02	Submit valid application	Fill all mandatory fields; click Submit	Application ID generated (LN-YYYY-NNNNNN); status = Submitted	Positive
UAT-02	US-01 / FR-01	Mandatory field validation	Leave Income blank; Submit	Error shown on field; submission blocked	Negative
UAT-03	US-01 / FR-03	Duplicate applicant detection	Enter PAN + mobile of an active application	Duplicate warning with link to existing record	Negative
UAT-04	US-02 / FR-01	Save draft and resume	Enter partial data; Save Draft; reopen	Draft saved; data intact; resumes at last step	Positive
UAT-05	US-03 / FR-04	e-KYC success	Enter valid PAN + Aadhaar consent	Status = KYC Verified; response stored	Positive

TC ID	Story / FR	Scenario	Test Steps (summary)	Expected Result	Type
UAT-06	US-03 / FR-04	e-KYC API timeout	Simulate vendor timeout	Retries 3 times; status = KYC Pending; alert raised	Negative
UAT-07	US-03 / FR-05	Document upload limits	Upload 7 MB file and an .exe file	Both rejected with clear message	Negative
UAT-08	US-04 / FR-07	FOIR breach flag	Applicant with FOIR 55%	Red Policy Breach flag; Approve disabled	Negative
UAT-09	US-05 / FR-08	Approval matrix routing	Submit ₹8L loan to Underwriter level	Auto-routed to Senior Underwriter	Positive
UAT-10	US-05 / FR-09	Reject without reason	Select Reject; leave reason blank	Action blocked; reason code mandatory	Negative
UAT-11	FR-10 / BRL-04	Straight-through approval	Score 780, FOIR 35%, ₹2L	Auto-approved; sanction letter generated	Positive
UAT-12	US-06 / FR-12	Maker-checker control	Maker tries to approve own disbursement	System denies; different checker required	Negative
UAT-13	US-06 / FR-12	Penny-drop failure	Use invalid bank account	Disbursement blocked; reason displayed	Negative
UAT-14	US-07 / FR-13	Payment allocation order	Post part-payment on overdue EMI with penalty	Allocated penalty → interest → principal; receipt generated	Positive
UAT-15	US-07 / FR-13	Excess payment	Pay more than EMI due	Excess reduces principal; schedule recalculated	Positive
UAT-16	US-08 / FR-15,16	Reminder and worklist	Run daily job at T-3 and T+3	SMS/email sent; worklist sorted by DPD and amount	Positive
UAT-17	FR-17	Foreclosure	Request foreclosure quote and pay	Correct amount computed; loan closed; NOC issued	Positive
UAT-18	US-09 / FR-18	Dashboard accuracy	Compare dashboard KPIs with SQL reconciliation	Variance = 0; filters work across visuals	Positive
UAT-19	US-10 / FR-19	Audit trail integrity	Edit an application; attempt to edit log	Change logged with old/new value; log edit denied	Security
UAT-20	NFR-01,03	Role-based access and performance	Login as RM; open Admin page; run 500-user load	Access denied; response < 3 sec (95th percentile)	Non-functional

11.3 Defect and Sign-Off

Stakeholder	Role in UAT	Sign-off
Head of Lending Ops	Process and functional acceptance	☐ Approved
Chief Risk Officer	Credit rules and approval matrix	☐ Approved
Compliance & Audit	KYC, audit trail, regulatory controls	☐ Approved
Finance	Disbursement and repayment accounting	☐ Approved
COO (Sponsor)	Go-live decision	☐ Approved

12. Requirements Traceability Matrix (Summary)

Objective	Business Req.	Functional Req.	User Story	UAT Case
BO-1, BO-2	BR-01	FR-01, 02, 03	US-01, US-02	UAT-01 to 04
BO-1, BO-7	BR-02	FR-04, 05	US-03	UAT-05 to 07
BO-1, BO-5	BR-03, BR-04	FR-06 to 09	US-04, US-05	UAT-08 to 11
BO-1	BR-05	FR-10, 11	US-05	UAT-11
BO-7	BR-06	FR-12	US-06	UAT-12, 13
BO-4	BR-07, BR-08	FR-13 to 16	US-07, US-08	UAT-14 to 16
BO-7	BR-09	FR-17	–	UAT-17
BO-6	BR-10	FR-18	US-09	UAT-18
BO-7	BR-11	FR-19	US-10	UAT-19, 20

13. Conclusion and Expected Outcomes

By digitising origination, standardising credit decisions, enforcing disbursement controls and giving management real-time visibility, the LMS is expected to cut TAT by about two-thirds, lift collection efficiency to 95%, bring PAR 30 down toward 8% and close audit gaps, while creating a scalable platform for future products and channels.